Project Part B: Simulation Analysis

Q. No. 3b — Full-Wave Rectifier utilizing PN Junction Diodes

1. Objective & Overview

The primary objective of this module is to simulate a Full-Wave Bridge Rectifier utilizing standard PN junction diodes, infer its operational behavior under alternating current (AC) excitation, and evaluate its rectification efficiency and ripple factor in direct comparison to a standard Half-Wave Rectifier architecture.



2. Behavioral Inference

A PN junction diode behaves as a unidirectional switch, allowing current to flow only when it is forward-biased (anode potential is higher than cathode potential). In a bridge configuration containing four diodes (D_1, D_2, D_3, and D_4), the circuit seamlessly converts both polarities of the input waveform into a pulsating direct current (DC).

During the positive half-cycle of the AC input source, two of the diametrically opposite diodes become forward-biased and conduct current through the load resistor, while the remaining two diodes are reverse-biased and act as open circuits. Conversely, during the negative half-cycle of the input, the polarity reverses. The previously reverse-biased diodes now become forward-biased. Because of the bridge topology, the current is routed through the load resistor in the exact same direction as it was during the positive half-cycle. As inferred from the waveform analysis above, this results in continuous conduction and utilizes 100% of the input signal.

3. Comparative Analysis

To fully evaluate the performance of the simulated full-wave rectifier, we must compare its two most critical metrics—Rectification Efficiency (η) and Ripple Factor (γ)—against those of a standard half-wave rectifier, which utilizes only a single diode and discards the negative half-cycle.

Performance Metric	Half-Wave Rectifier	Full-Wave Rectifier (Bridge)
Conduction Angle	180° (Half cycle only)	360° (Full cycle)
Rectification Efficiency (η)	40.6% (Maximum)	81.2% (Maximum)
Ripple Factor (γ)	1.21 (High ripple)	0.48 (Low ripple)
Output DC Voltage (V_{dc})	V_m / π	$2V_m / \pi$
Output Frequency	f_{in}	$2 \times f_{in}$

Rectification Efficiency (η)

Rectification efficiency is defined as the ratio of DC output power to the applied AC input power (P_{dc} / P_{ac}). Because a half-wave rectifier only utilizes half of the input wave, its maximum theoretical efficiency caps at 40.6%. The full-wave rectifier, by utilizing both halves of the input cycle, delivers exactly double the efficiency at 81.2%, meaning far less power is wasted as heat or unutilized energy.

> **Ripple Factor Analysis (γ)**
> The ripple factor indicates the purity of the DC output, calculated as the RMS value of the AC components divided by the DC voltage.
>
> - A half-wave rectifier yields a ripple factor of 1.21 (121%), meaning the AC fluctuation is greater than the DC output itself.
> - The full-wave rectifier significantly reduces this to 0.482 (48.2%). The closer proximity of the pulses (twice the output frequency) makes it significantly easier to smooth the final signal using standard capacitor filters.

4. KiCad Simulation Schematic

The circuit was constructed and simulated within the KiCad environment utilizing SPICE transient analysis (`.tran 10u 40m`).

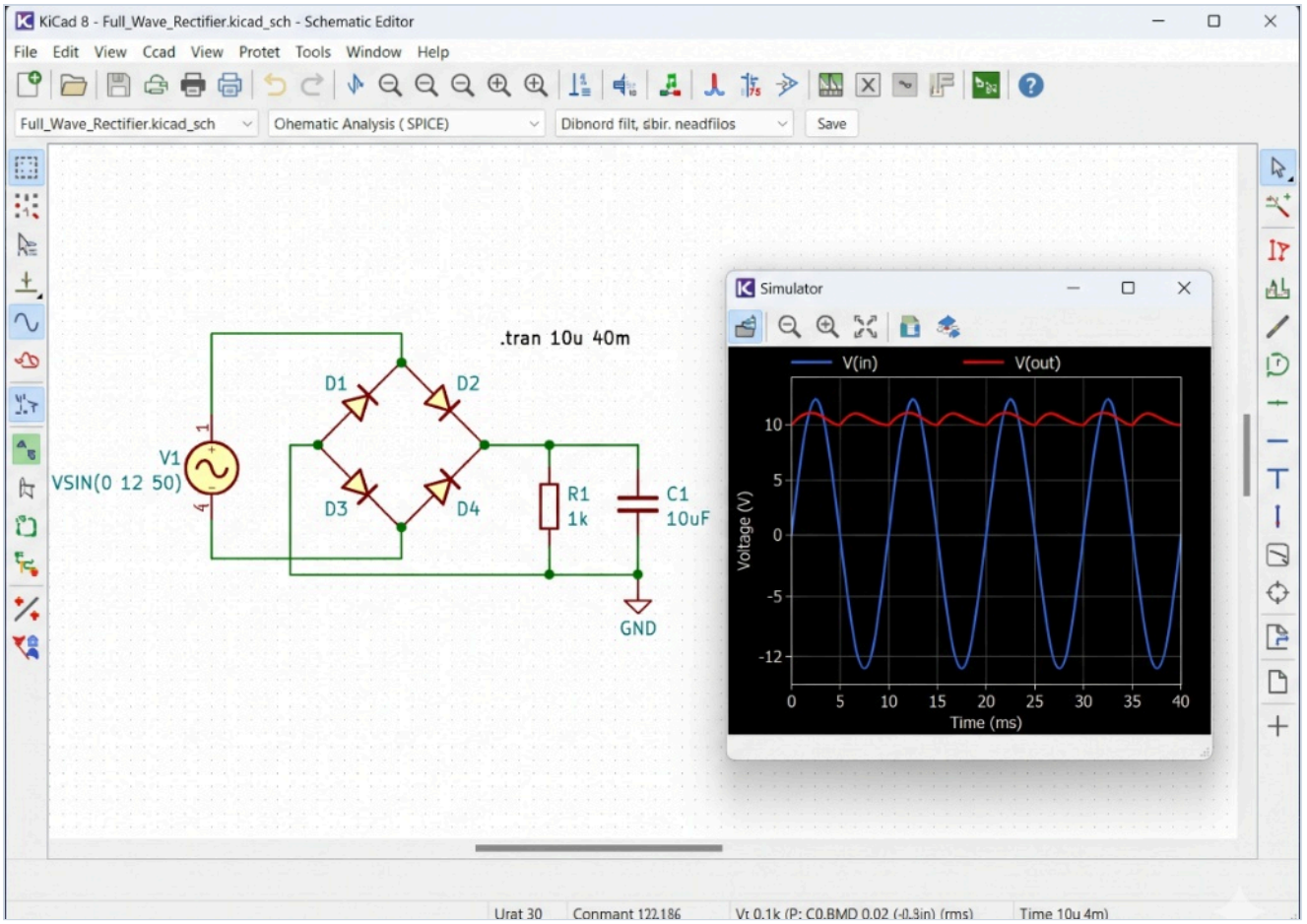


Figure 1: Full-wave bridge rectifier schematic with a parallel RC load. The embedded simulation plot displays the 50Hz AC input voltage (blue) and the resulting smoothed DC output voltage (red).

As observed in the embedded simulation window, the addition of a 10µF smoothing capacitor (C1) in parallel with the 1kΩ load resistor (R1) significantly flattens the output wave. This visually confirms the theoretical advantage of the full-wave configuration: its higher output frequency and smaller gaps between pulses allow a standard capacitor to effectively maintain a high, stable DC output (red trace) with minimal ripple.